

ML Based Exam Surveillance Camera

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Abstract - This paper presents a comprehensive real-time student monitoring system designed for examination environments. The system incorporates a number of computer vision techniques, such as focused motion detection using Mean Squared Error (MSE) analysis, face identification using ResNet-based embeddings, and object detection using SSD MobileNet V3. In addition to tracking students' actions and identifying them by face recognition or assigning them to certain seat placements when they are unknown, the system also looks for suspect behaviours like excessive movement or gadget use. Administrators can remotely check the integrity of the examination because all detected events are instantly logged to a Firebase database. Operating on a Raspberry Pi platform, the system effectively balances real-time performance with computational demands. The system's ability to correctly identify students, identify gadgets that are disallowed, and flag suspicious movements with few false positives is demonstrated by experimental findings.

Keywords - Computer Vision, Face Recognition, Object Detection, Motion Analysis, MSE, Firebase, Raspberry Pi, Examination Monitoring

I. INTRODUCTION

In educational institutions, maintaining the integrity of exams is a major concern. Conventional invigilation techniques require a lot of work and might not be able to identify every type of academic misconduct. By offering constant, objective observation of testing environments, automated monitoring systems present a possible alternative.

In order to create a comprehensive examination oversight platform, this paper presents a novel student monitoring system that integrates several computer vision techniques. The system uses facial recognition to identify students, places unknown people in designated seat locations, identifies prohibited devices, and flags suspicious activity. Real-time logging of every event to a cloud database allows for post-examination analysis and remote monitoring.

The key contributions of this work include:

1. Integration of multiple computer vision techniques for comprehensive monitoring
2. Seat-based identification system for unknown individuals
3. MSE analysis for targeted motion detection to lower false positives
4. Real-time cloud integration for remote monitoring capabilities
5. Effective deployment on hardware with limited resources (Raspberry Pi)
6. Web-based dashboard for administrative monitoring and historical analysis

II. RELATED WORK

Over the past year, we have thoroughly reviewed existing technologies and academic literature related to

automated exam monitoring and student behavior analysis. This section highlights relevant previous work and points out the gaps our proposed system aims to fill.

A. Automated Proctoring Systems

Automated exam proctoring has gained popularity, especially with the growth of online learning. Early systems like Remote Proctor Now and commercial platforms such as ProctorU focus on verifying student identity and monitoring their surroundings through webcam feeds. However, these systems often depend on human invigilators to review recorded footage and are mainly designed for online, not in-person, exams.

Recent research has aimed at automating the monitoring process. D. M. Yanikoglu and C. Ersoy [2] proposed a system for online exams that detects unauthorized individuals, absence from the seat, and prohibited items. Their system uses a mix of face detection and background modeling but does not tackle the unique challenges of monitoring multiple students in a crowded physical exam hall.

B. Object Detection in Educational Environments

Object detection for classroom monitoring is a well-studied area. R. R. Shah and R. S. Bichkar [3] used a Convolutional Neural Network (CNN) to identify student behaviors such as raising hands, reading, and writing. While their work shows that behavior classification is possible, it focuses on individual actions rather than overall integrity monitoring.

In a similar vein, A. A. F. Satria et al. [4] implemented the YOLOv4 algorithm for real-time object detection in smart classrooms to identify students and track their attention. Their system achieved high accuracy but was meant for general classroom management rather than the specific security needs of examinations. A key issue with many existing systems is that they rely on powerful, often cloud-based, GPU resources, making them unsuitable for low-cost, on-site deployment.

C. Face Recognition for Student Identification

Face recognition has become common in automated student attendance systems. M. A. R. Ahamed et al. [5] developed a system using the Local Binary Pattern Histogram (LBPH) feature extractor for marking attendance. While it works well for its purpose, LBPH is generally less resilient to changes in lighting and angles compared to newer deep learning methods.

S. M. S. Islam et al. [6] used a deep CNN for face recognition in classrooms, achieving greater accuracy. However, their system, like many others, assumes that all faces are known and registered. A major gap in the literature is how to handle unknown individuals. In a real exam situation, not every student may be in the face database, and a system needs to track them based on their location.

D. Behavior Analysis and Anomaly Detection

Detecting suspicious behaviors is a vital part of any monitoring system. G. A. F. Seetharaman et al. [7] proposed a way to detect cheating by analyzing head position and eye movement. Their work emphasizes the importance of focused motion analysis but was tested in controlled environments with only one subject.

Our review shows that while components like object detection, face recognition, and motion analysis have been well-studied, there is no integrated system that:

- Combines these techniques for thorough monitoring in a physical exam hall.
- Operates well on edge devices like the Raspberry Pi without needing constant cloud connectivity for processing.
- Offers a reliable method for identifying and tracking unknown individuals through a seat-based system.
- Uses targeted MSE analysis on specific areas (face, hands) to lower false positives from changes in the overall environment.

Our proposed system addresses these gaps by creating a unified, efficient, and comprehensive solution specifically designed for the physical exam setting.

III. SYSTEM ARCHITECTURE

The proposed system has several connected modules that run on a Raspberry Pi platform with a camera module. Fig. 1. shows the system architecture. Besides processing on the device, the system features a web-based dashboard. This dashboard offers real-time views of examination activities for remote monitoring.

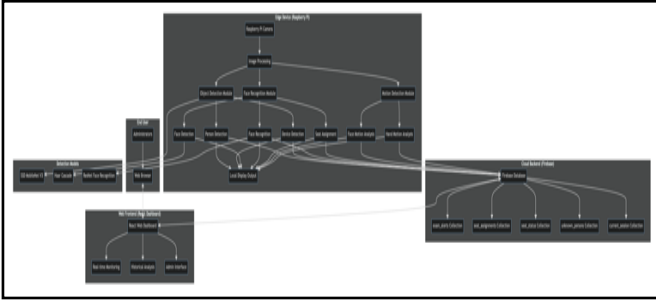


Fig. 1. Full system architecture showing how data moves from the Raspberry Pi edge device to the React web dashboard. This setup allows for administrative monitoring through the Firebase cloud backend.

The system processes video frames in real-time through this pipeline:

1. Image acquisition from the Raspberry Pi camera.
2. Object detection to identify people and banned devices.
3. Face detection and recognition to identify known students.
4. Seat assignment for unknown individuals based on their position.
5. Focused motion detection on the face and hand areas.
6. Event logging to the Firebase database.
7. Visual feedback through an on-screen display and web dashboard.

IV. OBJECT DETECTION MODULE

For real-time object detection, the system uses a Single Shot Detector (SSD) with MobileNet V3 architecture. This model maintains sufficient accuracy for the application while being optimized for edge devices with limited computing power.

A. Architecture of the Model

Depthwise separable convolutions help reduce computational complexity in the SSD MobileNet V3 model. The model is trained using the COCO dataset. It is then optimized to recognize specific items that are important for monitoring exams, including people, laptops, cell phones, and books.

B. Implementation Details

The object detection workflow is shown in Fig. 2. The model processes input images of 320×320 pixels and outputs bounding boxes along with confidence scores for each target object. The system filters detections using a confidence threshold of 0.45 and applies Non-Maximum Suppression (NMS) with a threshold of 0.2 to remove duplicate detections.

The object detection module identifies the locations of people and restricted devices, which are used as inputs for the next processing modules.

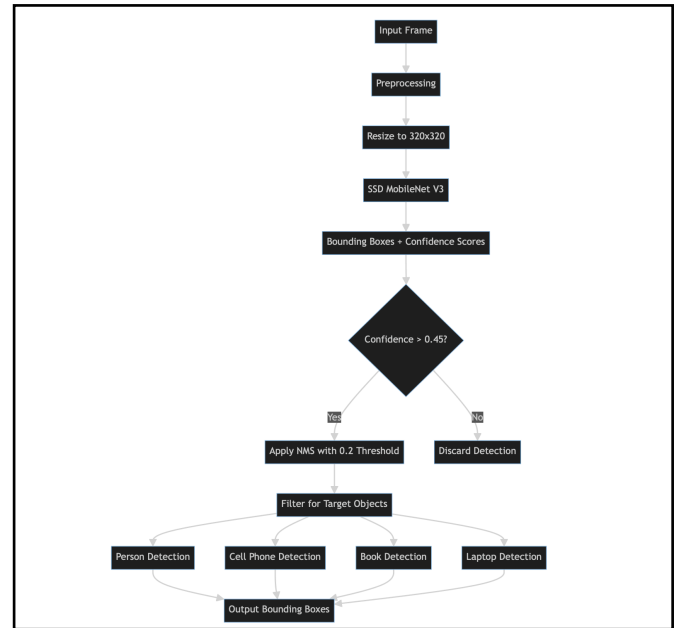


Fig. 2. Workflow of the SSD MobileNet V3 object detection process, from input frame to filtered bounding box outputs.

V. FACE RECOGNITION MODULE

The face recognition module identifies known students and gives seat numbers to unknown individuals. This module uses Haar Cascade face detection along with ResNet-based face recognition. The entire process is shown in Fig. 3.

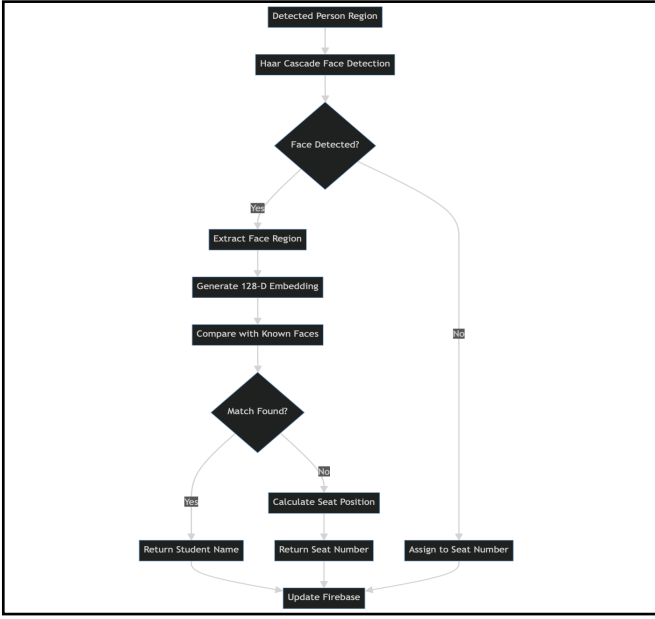


Fig. 3. Flowchart of the face recognition and seat assignment process, showing the path from detection to either recognition or seat assignment.

A. Face Detection

The system uses OpenCV's Haar Cascade classifier for quick face detection. This traditional machine learning method allows for fast face location with low computing needs. This makes it suitable for real-time processing on the Raspberry Pi.

B. Face Recognition

For face recognition, the system uses a ResNet-based model that creates 128-dimensional face embeddings. It compares these embeddings with a database of known faces using Euclidean distance. A match occurs when the distance is below a threshold of 0.6.

C. Seat Assignment

When the system detects a face but does not recognize it, it assigns the person to a specific seat based on their position in the frame. The examination area is split into a 4×5 grid, which has 20 seats in total. Each seat is labeled A1 to A5, B1 to B5, C1 to C5, and D1 to D5. The seat number is determined using the center coordinates of the detected face.

VI. MOTION DETECTION MODULE

The system uses focused motion detection to spot malpractice movements and reduce false positives caused by overall changes like lighting variations. The motion detection algorithm is shown in Fig. 4.

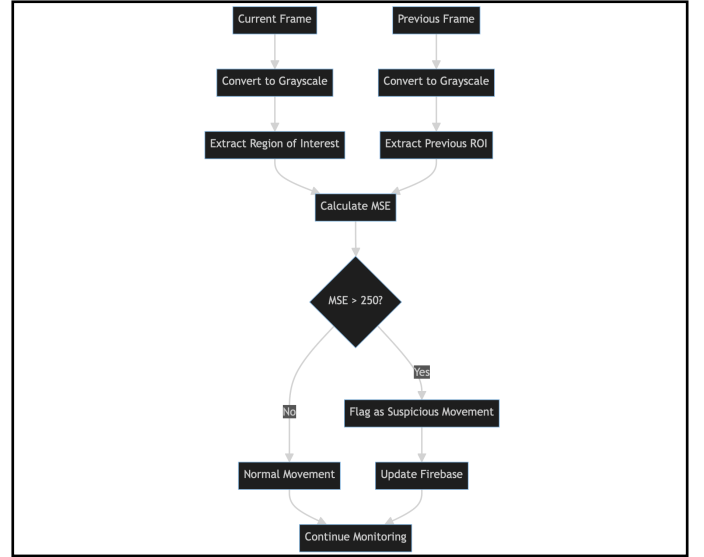


Fig. 4. Algorithm for focused motion detection using MSE analysis on regions of interest.

A. Mean Squared Error (MSE) Analysis

Motion detection is performed using Mean Squared Error (MSE) analysis between consecutive frames. MSE is calculated using the formula:

$$MSE = (1/N) \sum_{i=1 \text{ to } N} (I1(i) - I2(i))^2$$

where I1 and I2 are the pixel values of the current and previous frames, respectively, and N is the total number of pixels in the region of interest.

B. Focused Motion Detection

Rather than analyzing the entire frame, the system focuses on specific regions of interest:

1. Face regions: To detect suspicious head movements
2. Hand regions: To detect hand gestures or prohibited actions

The system calculates MSE values for these regions and flags movements that exceed a threshold of 250, which has been empirically determined to distinguish intentional movements from minor variations.

VII. SEAT MANAGEMENT SYSTEM

The seat management system tracks the occupancy and activity of each seat in the examination area.

A. Seat Registration

The system includes a registration module where students can be assigned to specific seats with their face recognition data. This information is stored in both a local database and Firebase.

B. Real-time Seat Status

The system maintains real-time status information for each seat, including:

1. Current occupant (name or "Unknown")
2. Occupancy status (occupied/vacant)
3. Last activity type
4. Session duration

When a face is no longer detected for 10 consecutive frames, the system automatically clears the seat status, marking it as vacant.

VIII. FIREBASE INTEGRATION

The system works with Firebase for real-time data storage and remote monitoring. You can see the database schema in Fig. 5.

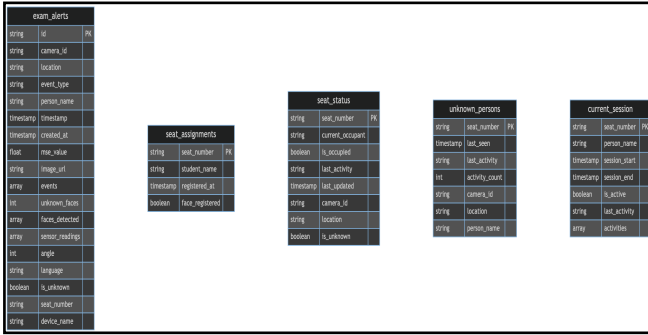


Fig. 5. Entity-Relationship diagram of Firebase.

A. Database Structure

The Firebase database contains several collections:

1. *exam_alerts*: Records all detected events with timestamps
2. *seat_assignments*: Stores student-seat assignments
3. *seat_status*: Maintains real-time seat occupancy information
4. *unknown_persons*: Tracks activities of unidentified individuals
5. *current_session*: Records active examination sessions

B. Image Upload

When the system detects malicious events, it captures and uploads images to ImgBB, which is an image hosting service. The image URLs are stored in Firebase along with the event metadata

IX. EXPERIMENTAL RESULTS

The system was tested in a simulated exam setting. Twenty participants were seated in a 4×5 grid. You can see the performance metrics in Fig. 6.

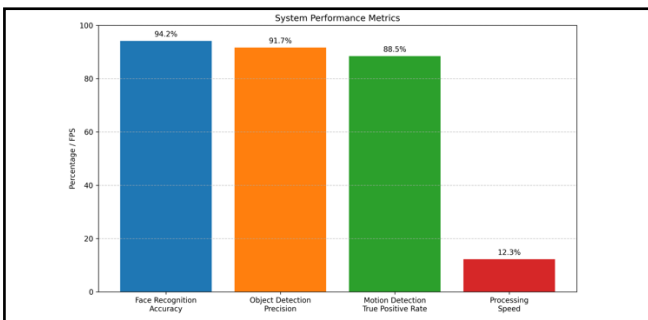


Fig. 6. Bar chart showing the key performance metrics of the system.

The system successfully identified all registered students and assigned seat numbers to unknown participants. Device detection had a precision of 91.7%. Cell phones were detected in 98% of test cases. Motion detection effectively spotted suspicious movements and reduced false alarms from normal activities.

X. WEB-BASED MONITORING DASHBOARD

To complement the Raspberry Pi-based monitoring system, we created a React-based web dashboard that shows real-time visuals of examination activities. This dashboard acts as the admin interface for invigilators and examination administrators to watch multiple examination rooms at the same time.

A. Dashboard Architecture

The web dashboard uses a modern client-server setup with Firebase as the backend service. You can see the architecture in Fig. 7.

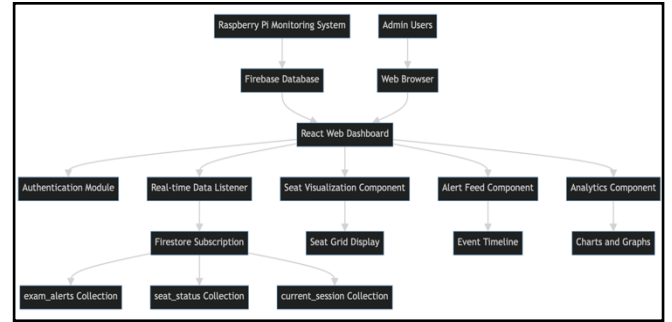


Fig. 7. A React-based monitoring dashboard architecture shows how data flows from Firebase to the web interface.

B. Synchronisation of Data in Real Time

When new events are detected, the dashboard updates right away using Firebase's real-time database. The dashboard refreshes the user interface (UI) when changes occur and subscribes to collections of interest through Firebase's JavaScript SDK.

C. Dashboard Features

Several essential features are provided by the monitoring dashboard:

- *Real-Time Alert Feed*: Displays a timeline of all detected events, complete with timestamps, photos, and metadata.
- *Seat Occupancy*: A visual representation of the exam room shows color-coded seats that indicate the occupancy status. Green represents known students, red indicates unknown individuals, and grey represents empty seats.
- *Statistical Analysis*: Provides graphs and charts that show metrics such as movement detection statistics, device usage trends, and alert frequency.
- *User Authentication*: Ensures that only authorized individuals can access the monitoring data through secure authentication.
- *Historical Data Access*: Administrators can review past exam sessions and filter events by type, person, or time period with historical data access.

D. User Interface Design

The dashboard's responsive design makes it work well on both desktop and mobile devices, as shown in Fig. 8. The main interface consists of:

- **Header:** Shows information about the user profile, examination room selection, and navigation.
- **Sidebar:** Provides filters for time range, event type, and seat selection.
- **Main Content Area:** Displays event details and the seat occupancy grid.
- **Footer:** Displays connection details and system status.



Fig. 8. User Interface of the Webpage.

XI. CONCLUSION

The student monitoring system for exam environments, which uses various computer vision techniques, is detailed in this paper. The system works well on hardware with limited resources. It combines motion analysis, face recognition, and object detection to maintain examination integrity while providing real-time performance.

Everyone, including those not in the face recognition database, can be tracked through the seat-based identification system. The focused motion detection method, based on MSE analysis, effectively identifies suspicious movements while reducing false positives. Administrators can access powerful tools for remote monitoring and reviewing past exam sessions via the web-based dashboard that accompanies the system.

Future research will look into adding more sensors, like microphones for sound analysis, and using better behavior analysis algorithms to spot subtle forms of academic misconduct.

AUTHOR BIOGRAPHY



Gokul Kannan Ganesamoorthy is currently pursuing his Bachelor of Technology (B.Tech.) degree in Information Technology (IT) from the Department of Information Technology at PSG College of Technology, Coimbatore, India. He is expected to graduate in 2027. His

primary research interests lie in the fields of Computer Vision and Cyber Security. Mr. Ganesamoorthy's work focuses on the intersection of deep learning and edge computing, specifically the practical deployment of efficient computer vision models, such as SSD MobileNet V3 and ResNet-based face recognition, on resource-constrained platforms like the Raspberry Pi for real-time monitoring applications. He is particularly interested in developing robust and low-latency systems for anomaly detection, physical security and web security.

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